Study design: freeze both programs, then test the same scores

IDENTITY_CORE_v1 (21 genes)
INS, PDX1, MAFA, NKX6-1, SLC30A8…
Locked before any T2D contrast

STRESS_CORE_v1 (18 genes)
DDIT3, ATF4, XBP1, HSPA5, TXNIP…
Prespecified direction: up in T2D / hyperglycemia

Both scores applied to every cohort below

Discovery
GSE50244 RNA-seq
n=89 islets
HbA1c endpoint
(n=77 complete)

External
validation
GSE76894 array
19 T2D / 84 ND
age/sex/BMI/batch

Same-biobank
companion
GSE38642 array
53/89 donor overlap
not independent

Orthogonal LCM
GSE20966
10 vs 10
underpowered check

Primary family (4 tests)
Identity and stress on:
• GSE50244 score ~ HbA1c + age + sex + BMI
• GSE76894 score ~ T2D + age + sex + BMI + batch
No gene added from differential expression.
Stress-up is allowed to fail.

Overlays, not extra endpoints
GTEx v8 public median TPM:
tissue selectivity (bulk pancreas, not islets)
Open Targets T2D overlay (not MR)
NHANES / MIMIC / WGCNA / two-sample MR excluded